VibeVisor: Formal Specifications for a Production-Grade Implementation Blueprint

Designated Hardware Environment: Apple Silicon Architecture (M1/M2/M3)
Systemic Architecture: Rust-Based QEMU Orchestration Framework
Interface Substrate: Tauri Integration with Svelte 5 (Runes) and Tailwind CSS
Backend Infrastructure: Rust (Tokio) with an Optimized QEMU 9.0+ Curated Backend

I. Executive Abstract

The entity designated as VibeVisor represents a high-performance, memory-secure virtualization apparatus engineered for the macOS environment. In contradistinction to rudimentary wrapper implementations, a methodology involving a curtailed backend architecture and a Rust-based control logic is employed. This approach is intended to facilitate a professional, high-velocity user experience characterized by a significantly diminished computational footprint relative to existing incumbent solutions, specifically the UTM framework.

II. Preparatory Phase: Toolchain Synthesis and Backend Minimization

Mandatory Objective: The systematic eradication of superfluous computational overhead via the construction of a curtailed QEMU binary.

- **Strategic Compilation Methodology:** * The QEMU source repository is to be cloned, with configuration parameters restricted to the aarch64-softmmu target.
    - There shall be a comprehensive deactivation of non-essential legacy subsystems, including but not limited to: audio, vnc, sdl, gtk, oss, parallel, and fpga.
    - High-velocity features—specifically hvf (Hypervisor.framework), spice, virtfs, and libusb—are to be explicitly enabled.
- **Artifact Archival and Management:** Upon successful compilation, the resulting binary shall be situated within the src-tauri/sidecars directory to facilitate subsequent bundling procedures.
- **Security Entitlements:** A vibe.entitlements documentation shall be established, incorporating the com.apple.security.hypervisor and com.apple.vm.networking cryptographic keys.

III. Primary Phase: Development of the Central Logic Component (vibe-core)

Mandatory Objective: The establishment of a memory-secure computational core.

- **The vibe-launcher Subsystem:** * Asynchronous process oversight is to be managed via the tokio::process library.
    - An argument synthesis engine shall be developed to generate optimized parameters for the Apple Silicon environment (notably -accel hvf and -cpu host).
- **The vibe-qmp Subsystem:** * Strongly-typed Rust abstractions for the QEMU Machine Protocol are to be instantiated.
    - The utilization of the serde framework for type-safe JSON-RPC communication across Unix Domain Sockets is mandated.
    - A real-time event distribution bus shall be maintained to monitor transitions in virtual machine states (e.g., Initialization, Termination, Critical Failure, or Suspension).

IV. Secondary Phase: Optimization of Input/Output and Network Topologies

Mandatory Objective: The reduction of latency and the maximization of data throughput.

- **Data Storage Management:** * Management of .qcow2 disk images shall be executed through Rust-based implementations.
    - Functionality pertaining to state snapshots and "Linked Clones" is to be integrated for the purpose of preserving non-volatile storage space.
- **Network Topology:** * Support for the vmnet.framework is to be integrated to facilitate shared or bridged network configurations.
    - The resolution of privilege escalation requirements for network interface initialization shall be addressed through a specialized Rust-based utility.
- **File System Integration:** * The virtio-fs protocol is to be configured to ensure near-native data transfer velocities between the host macOS and the guest Linux environments.

V. Tertiary Phase: Implementation of the Refined Graphical Interface (Svelte 5 and Tauri)

Mandatory Objective: The provision of an interface characterized by a refined, professional-grade aesthetic consistent with contemporary macOS standards.

- **Architectural Design Tokens:**
    - **Visual Aesthetic:** The utilization of Gaussian blur substrates (backdrop-blur), SF Pro Rounded typography, and a chromatic palette prioritized for diminished light environments (Dark Mode) is required.
    - **State Management:** Reactive and high-performance interface updates shall be governed by the Svelte 5 "Runes" methodology ($state, $derived, $effect).
- **Functional Attributes:**
    - **Command Interface:** A Cmd+K palette is to be instantiated for the immediate administration of virtual machine entities.

    - **Visualization Engine:** A high-performance SPICE client shall be integrated, featuring automated resolution synchronization.
    - **Analytical Monitoring:** Real-time visual representations of central processing unit and random-access memory utilization are to be displayed for each virtual machine instance.

VI. Quaternary Phase: Integration of Specialized Functional Attributes

Mandatory Objective: The implementation of advanced features intended to establish a distinct technical superiority.

- **Direct Kernel Execution:** The circumvention of UEFI/BIOS initialization stages is sought to facilitate Linux kernel activation in a temporal span of less than two seconds.
- **Automated Hardware Optimization:** Logic shall be implemented to identify and allocate Performance (P-cores) and Efficiency (E-cores) cores pursuant to the immediate computational demands of the virtual machine.
- **Rapid State Preservation:** Rust-governed preservation of volatile memory states is to be established, facilitating near-instantaneous resumption of virtual machine operations.

VII. Final Phase: Dissemination and Systematic Optimization

Mandatory Objective: The preparation of a commercially viable distribution package accompanied by rigorous security protocols.

- **Computational Optimization:** Link Time Optimization (LTO) shall be applied to the Rust-based binaries, alongside the systematic removal of non-essential symbols from the QEMU sidecar.
- **Security Verification:** An automated pipeline (GitHub Actions) is to be utilized for the execution of cryptographic signing and Apple Notarization procedures.
- **Distribution Apparatus:** A curated .dmg or .pkg installer shall be produced, incorporating all requisite ancillary binaries.

VIII. Critical Assessment of Comparative Technical Advantages

Functional Attribute	**UTM (Incumbent Solution)**	**VibeVisor (Proposed Entity)**
Orchestration Logic	Swift / Objective-C	**Contemporary Rust (Asynchronous/Secure)**
Backend Footprint	Generic QEMU (~500MB)	**Curated Backend (~120MB)**
Interface Framework	SwiftUI	**Svelte 5 (Compiler-Optimized)**
Activation Velocity	Standard	**Accelerated (Direct Kernel**

	(Firmware-Dependent)	**Execution)**
Interaction Paradigm	General Utility	**Refined Professional Aesthetic**

IX. Directives for Automated Logic Synthesis

1. **Initiation of Primary Logic:** "It is requested that a Rust module, designated as vibe-launcher, be constructed to facilitate the instantiation of a QEMU process as a subordinate entity. The tokio::process::Command library must be utilized to synthesize parameters for an aarch64 Linux guest environment characterized by 4GB of random-access memory and HVF acceleration."
2. **Establishment of Inter-Process Communication:** "The instantiation of a Rust-based QmpClient is required. This client shall establish a connection via a Unix Domain Socket to transmit a query-status directive, subsequently translating the hexadecimal response into a strongly-typed Rust data structure via the Serde framework."
3. **Synthesis of the Graphical Substrate:** "A framework involving Tauri and Svelte 5 is to be established. A lateral navigation bar incorporating Gaussian blur elements is requested, alongside the implementation of a virtual machine registry that dynamically updates status indicators pursuant to events emitted by the Rust-based backend infrastructure."